Project Planning & Management

E-commerce Supply Chain | Data Engineering Project

Project Status: In Progress | Expected Completion: 4 Weeks

1. Project Overview

The E-commerce Supply Chain Data Engineering project is currently in progress. The team is developing a data-processing workflow to prepare, transform, store, and validate e-commerce supply-chain data so that it can support reliable analysis and reporting. The team expects to complete the project within the next four weeks. The dataset has been prepared/selected, while the remaining implementation, integration, testing, documentation, and presentation activities are planned across the schedule below.

2. Objectives

- Build a repeatable data pipeline for the selected e-commerce supply-chain dataset.
- Use Python and Pandas for data processing and transformation.
- Use SQL and PostgreSQL to query and store structured data.
- Use Apache Airflow to orchestrate and schedule pipeline tasks.
- Use Power BI to present relevant findings and operational metrics.
- Validate data and pipeline outputs and document the workflow for reproducibility.

3. Technology Stack

Technology	Planned role in the project
Python	Core programming language for pipeline logic and data processing.
Pandas	Data cleaning, transformation, profiling, and preparation.
SQL	Queries, aggregation, and validation of structured data.
PostgreSQL	Relational storage for processed data and analytical tables.
Apache Airflow	DAG-based orchestration and scheduling of pipeline tasks.
Power BI	Dashboards and reporting using metrics supported by the dataset.
Git/GitHub	Version control, collaboration, and project code management.
Pytest	Testing selected Python functions and data-processing rules.
Docker (optional)	Reproducible local environment if time and setup allow.

4. Scope

In scope

- Dataset profiling and documentation of relevant fields.
- Data cleaning and transformation using Python/Pandas.
- Loading and querying structured data in PostgreSQL using SQL.
- Airflow orchestration for the agreed pipeline tasks.
- Data validation and testing.
- Power BI reporting based on fields and metrics available in the dataset.
- Documentation, final review, and presentation.

Out of scope unless separately approved

- Production deployment for a real company.
- Live integration with commercial e-commerce systems.
- Claims about real-time processing, inventory optimization, or prediction unless implemented and evaluated.
- Any results or performance improvements that have not been measured.

5. Four-Week Schedule

Week	Planned activities	Expected deliverables
Week 1	Confirm requirements and business questions; review literature; finalize schema and architecture; inspect data quality.	Requirements document, literature review, data dictionary, architecture outline.
Week 2	Implement/refine Python and Pandas transformations; prepare PostgreSQL schema and SQL queries; start Airflow DAG.	Initial pipeline stages, target tables, first orchestration workflow.
Week 3	Integrate and run the pipeline through Airflow; implement quality checks and tests; connect processed data to Power BI.	Integrated pipeline, test/validation evidence, initial dashboard.
Week 4	Resolve issues; validate final outputs; finalize report, technical documentation, and presentation/demo.	Reviewed project package, final dashboard/report, presentation and acceptance checklist.

6. Work Breakdown Structure

- Project coordination: maintain scope, schedule, decisions, and blockers.
- Requirements gathering: define users, business questions, data needs, and acceptance criteria.
- Literature review: review e-commerce analytics, supply-chain analytics, ETL, data quality, and orchestration.
- Data understanding: document source, schema, data types, quality issues, and field limitations.
- Pipeline development: implement the agreed Python/Pandas transformations and PostgreSQL loading/query steps.
- Orchestration: create and test Airflow DAGs for the selected tasks.
- Validation: apply data-quality checks and tests; investigate unexpected outputs.
- Analytics and closure: build Power BI outputs, document limitations, prepare final presentation.

7. Roles and Responsibilities

The role allocation below is a proposed starting point and should be adjusted to match the team's actual division of work.

Member	Proposed focus
Menna Hamdy	Project coordination, requirements, and documentation tracking
Omar Salem	Data profiling and data preparation
Ibrahim Ali	Python/Pandas pipeline implementation
Belal Yasser	SQL/PostgreSQL and data validation
Ahmed Sami	Airflow orchestration and integration
Abdalla Elhelaly	Power BI reporting, presentation, and final review

8. Monitoring and Communication

- Hold a weekly progress meeting and review the next milestone.
- Track each task with an owner, due date, status, and blocker.
- Use Git/GitHub for version control and collaboration as agreed by the team.
- Maintain an issue log and record decisions about data definitions and transformations.

9. Risk Register

Risk	Potential impact	Mitigation
Airflow environment or dependency issues	Pipeline scheduling is delayed	Set up a minimal local workflow early and test one simple DAG before adding complexity.
Data quality or unclear field meanings	Incorrect transformation or reporting	Profile the data and document assumptions; validate rules against actual fields.
PostgreSQL integration issues	Data loading or querying fails	Test connectivity and table schema early in Week 2.
Scope growth	Four-week deadline at risk	Prioritize the minimum viable pipeline and defer optional Docker/advanced features.
Unsupported business claims	Report credibility risk	Only report metrics supported by available fields and measured tests.
Team integration delays	Late end-to-end testing	Integrate components incrementally and assign clear owners.

10. Completion Criteria

- The agreed pipeline can be run using documented steps.
- Python/Pandas processing, PostgreSQL storage/querying, and Airflow orchestration work together as intended.
- Relevant data-quality checks and test outcomes are documented.
- Power BI visuals, if included in the final scope, use documented metrics from the processed data.
- Known limitations are disclosed and all project documents are reviewed before submission.

11. Planning Assumptions

This plan assumes that the project is currently being developed and is expected to be completed in four weeks. Python, Pandas, SQL, PostgreSQL, Power BI, and Apache Airflow are treated as core project technologies based on the team's confirmation. Pytest is included for testing, while Docker remains optional. The schedule is a working plan and should be updated to reflect completed tasks and any changes in team availability.

Project Team

- Menna Hamdy
- Omar Salem
- Ibrahim Ali
- Belal Yasser
- Ahmed Sami
- Abdalla Elhelaly